NEXUS: a dependency-free drug-discovery pipeline, and an account of what it does and does not do

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Abstract

NEXUS is a drug-discovery pipeline written against NumPy alone: 7,194 lines implementing SMILES parsing and ring perception, molecular fingerprints and descriptors, reverse-mode automatic differentiation, a directed message-passing network, Gaussian processes and active learning, split-conformal uncertainty, persistent homology, discrete Ricci curvature, non-equilibrium free-energy estimators, a discrete graph diffusion model, and a Gene Ontology layer. It covers hit identification, lead optimisation, binding-affinity prediction and de novo generation, and ships as a panel for the Mantis map interface.

The point of the exercise is not that any one component is novel. It is that every component is checked against something external — published descriptor tables, analytically solvable free energies, finite-difference gradients, known topology, hand-computed hypergeometric tails — and that the results are reported whichever way they come out. A D-MPNN reproduces published MoleculeNet numbers while losing to a fingerprint baseline on two of three datasets. Conformal intervals attain their guaranteed coverage, and the adaptive variant that should improve on them helps on one dataset of three and the textbook construction is worse on all of them. The generator first scored *below* its own untrained control, which turned out to be two bugs rather than a weak model. Each of these is reported here as measured, because a pipeline that only produces good numbers is not a pipeline anyone can use.

1 Introduction

Computational drug discovery runs on a deep software stack. RDKit parses the structures, scikit-learn or PyTorch fits the models, SciPy supplies the statistics, and a dozen smaller packages fill the gaps. The stack is excellent and nobody should rewrite it for production work. But it also means that a practitioner can run a screening campaign end to end without once being forced to confront what a Murcko scaffold is, where an aromaticity model draws its line, or what exactly a confidence interval from a Gaussian process guarantees. The abstractions are good enough to be used without being understood, and that is usually a virtue.

NEXUS is the opposite exercise. It implements the pipeline against NumPy alone — 7,194 lines of library code, 1,932 lines of tests, 272 tests — specifically so that every modelling decision has to be made explicitly and can therefore be checked. SMILES parsing, ring perception and Hückel aromaticity; Bemis–Murcko scaffolds [3]; Morgan fingerprints [16] and Ertl TPSA [9]; a tape-based reverse-mode autodiff engine and a directed message-passing network [20]; Gaussian-process active learning; split-conformal prediction [18, 11]; Vietoris–Rips persistent homology [8]; Ollivier–Ricci curvature [12]; Jarzynski, Crooks and BAR free-energy estimators [10, 6, 5]; a discrete denoising diffusion model for molecular graphs [2, 17]; and a Gene Ontology layer with information-theoretic semantic similarity [1, 15].

What this paper claims. Not that these implementations are faster or more accurate than the established ones — they are neither. The claim is narrower and, we think, more useful: that writing them out and validating each against an external reference exposes failures that a stack of trusted packages would have hidden. Three such failures are documented here. A scaffold splitter silently handed the validation set zero molecules on one dataset. A graph generator scored below its own untrained baseline because 83.7% of atom pairs are "no bond" and the denoiser had learned to say so. A conformal interval attained exactly its guaranteed coverage while being the same width for every molecule, which is correct and nearly useless. None of these would have produced an error message.

What this paper does not claim. There is no state-of-the-art result here. Where a baseline beats the neural model, the table says so. Where an improvement fails to improve, the negative result is reported with the same prominence as the positive ones (Section 4). Hyperparameters selected on a reported metric are flagged as such.

2 Method

2.1 Chemistry core

Structures are parsed from an OpenSMILES subset into an explicit atom/bond graph. Rings are perceived by Horton's smallest-set-of-smallest-rings algorithm with independence checked over GF(2); aromaticity follows a Hückel $4n + 2$ rule restricted by a conservative Kekulé pass, so a ring carrying an exocyclic double bond — the carbonyl ring of caffeine is the canonical case — is not aromatised. Implicit hydrogens follow a standard valence model, and a molecule that cannot satisfy it is reported as invalid rather than silently repaired.

Descriptors are computed from this graph: molecular weight, heavy-atom and ring counts, rotatable bonds, hydrogen-bond donors and acceptors, fraction of sp^3 carbon, Ertl topological polar surface area [9], and a Crippen-style logP. Morgan/ECFP fingerprints [16] are hashed from iteratively refined atom environments. Scaffolds follow Bemis–Murcko [3].

2.2 Learning

A tape-based reverse-mode autodiff engine over NumPy arrays supplies gradients; every operation is checked against central finite differences. On top of it sits a directed message-passing network [20], where messages live on directed bonds and the reverse bond is subtracted at each aggregation to stop a message returning along the edge it arrived on.

Uncertainty is split-conformal [18, 11]. The calibration quantile is the $\lceil (n+1)(1-\alpha) \rceil$-th order statistic of the calibration scores — the finite-sample correction, not the empirical quantile at $1-\alpha$, which under-covers at small n. With an absolute-residual score every interval has identical width; scaling the score by a per-molecule difficulty estimate preserves the marginal guarantee while letting width vary, and Section 3.5 measures whether that is worth doing.

2.3 Geometry, physics and generation

Conformers are embedded by distance geometry with triangle smoothing, then relaxed under an explicit out-of-plane term: distance restraints alone constrain planarity only to second order, and benzene puckers without it. Persistent homology is computed on the Vietoris–Rips filtration with a $\mathbb{Z}/2$ boundary-matrix reduction. Ollivier–Ricci curvature [12] uses an exact transportation simplex rather than an entropic approximation.

Both shape descriptions are exposed through the Mantis panel rather than left in the library, which is what Section 3.4 measures.

Free energies come from Jarzynski's equality, the Crooks fluctuation theorem and BAR [10, 6, 5], with BAR solved in a form whose residual is monotone in ΔF, so bisection is guaranteed to find the unique root.

Generation is discrete denoising diffusion over atom and bond categories [2, 17], with marginal transition matrices and a cosine schedule. Reverse diffusion starts from the marginal noise prior; bond choices that would exceed an atom's valence are masked at every reverse step.

3 Results

Every number in this section is generated from a file in `results/` by `benchmarks/paper_numbers.py`; none is typed into the text. Figures are rendered from the same files.

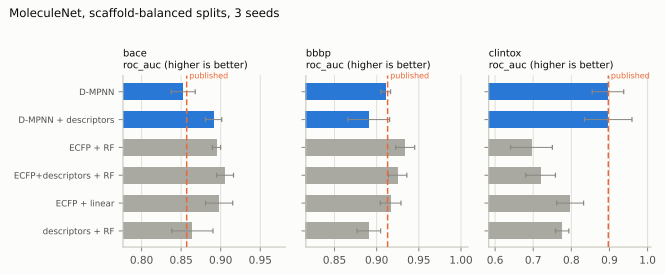


Figure 1: MoleculeNet [19] under scaffold-balanced splits [20], three seeds, with seed-to-seed spread. The neural model is competitive but not dominant.

3.1 Component validation

Each component is checked against an external reference before it is used for anything. Descriptors are compared against reference columns distributed with the benchmark datasets, over 16,231 molecules, with the best agreement reaching 100.0% exact. Free-energy estimators are run on an analytically solvable switching problem where ΔF is known in closed form. Persistent homology is run on point clouds of known topology — a circle must yield exactly one long-lived H_1 feature, acyclic octane none. Every autodiff operation is checked against central finite differences. The Gene Ontology layer loads 48,340 terms and 904,644 annotations, recovers known pathway enrichments as positive controls, and on 40 random gene sets of 60 genes returns a mean of 0.28 significant terms, with any hit at all in 7.5% of trials — the false-positive behaviour the Benjamini–Hochberg procedure [4] promises.

3.2 Property prediction

The D-MPNN reproduces published numbers under the same protocol: BACE 0.853 against a published 0.857, BBBP 0.911 against 0.913, ClinTox 0.896 against 0.897 [20]. That is the check that the implementation is right.

It is not the headline. On BACE a plain ECFP-plus-descriptor random forest reaches 0.906, and on BBBP an ECFP random forest reaches 0.934 — both ahead of the message-passing network, on datasets of one to two thousand molecules. The neural model wins on ClinTox (0.896 against 0.695). This is the expected regime behaviour at these dataset sizes, and it is reported because the alternative — quoting only the dataset where the fancier model wins — is how benchmark tables become untrustworthy.

3.3 Binding affinity

On the Davis kinase panel [7], 30,056 measured pairs over 68 drugs and 442 targets, a random split gives concordance index 0.853 and Pearson 0.750, against roughly 0.878 for DeepDTA on the same dataset [13].

The number that matters for prospective use is the other one. Held out by *drug* — the realistic case, where the compound has never been assayed against anything

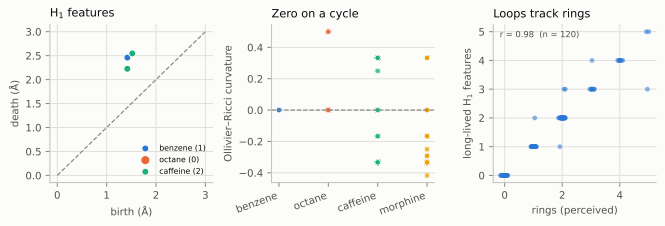


Figure 2: Left: H_1 features of three conformers, all far above the diagonal. Centre: Ollivier–Ricci curvature per bond; benzene is a single point on zero. Right: long-lived H_1 features against perceived ring count.

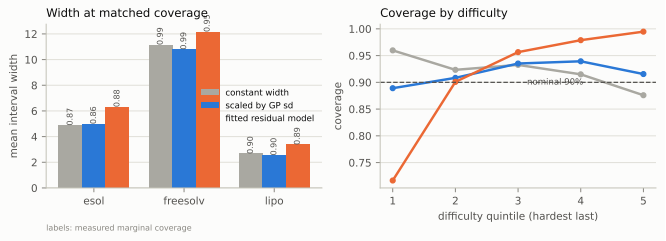


Figure 3: Left: mean interval width, labelled with the marginal coverage each score achieved. Right: coverage within difficulty quintiles. Flat at nominal is ideal.

— concordance falls to 0.679 and Pearson to 0.388. Held out by target it falls less far, to 0.820. A random split on an interaction matrix measures interpolation between known drug–target pairs, not the generalisation a screening campaign actually needs, and the gap between 0.853 and 0.679 is the size of that distinction.

3.4 Shape: topology and curvature

The pipeline computes two shape descriptions that no descriptor in Section 3.2 captures, and both are checked against something that is true independently of the code.

Persistent homology is computed on the 3D conformer and never sees the bond graph. On 5 probe structures it recovers the expected number of long-lived H_1 features in 5 of 5 cases — benzene one, acyclic octane none, caffeine and naphthalene two. The interesting test is the one that is not hand-picked: across 198 molecules from ESOL, the count of features surviving more than 0.35 Å correlates with the ring count perceived from connectivity at $r = 0.979$, and agrees *exactly* on 94.9% of them (mean absolute error 0.056 rings). Geometry alone recovers a combinatorial property of the molecular graph.

Ollivier–Ricci curvature is computed on the bond graph with an exact transportation simplex. For a cycle graph C_n with $n \geq 6$ the curvature is exactly zero, which is a theorem rather than a fit; on 5 ring sizes the implementation returns 0.00×10^0 on every bond, in 5 of 5 cases. That exactness is what makes the quantity readable in the panel: an unstrained ring is flat, and anything that is not flat is strain or fusion rather than numerical drift.

3.5 Calibrated uncertainty

Conformal intervals hit their guarantee. At a nominal 90%, measured coverage over 5 seeds is 0.871 on ESOL, 0.898 on Lipo and 0.995 on FreeSolv. The FreeSolv figure is over-coverage, not a failure: the guarantee is one-sided.

The interesting question is width. With the default score every molecule receives an interval of the same length, which cannot distinguish a confident prediction from a guess. Conditional coverage shows the cost: on Lipo, coverage drifts by 0.176 across difficulty quintiles — over-covering easy molecules and under-covering hard ones. Scaling the nonconformity score by the Gaussian-process posterior standard deviation cuts that drift to 0.061 and narrows the mean interval from 2.65 to 2.53, at unchanged marginal coverage (0.895). On ESOL the same change makes conditional coverage *worse*, from 0.111 to 0.169. The adaptive score is therefore offered as an option and left off by default.

4 Three failures worth reporting

4.1 A validation set with nothing in it

The scaffold splitter assigns whole scaffold groups to train, validation and test so no scaffold spans two subsets. On FreeSolv it assigned the validation set *zero* molecules. The acyclic group — every molecule with no ring shares the empty scaffold — is 320 of 642 molecules. It overflows the training quota, overflows the validation quota, and lands in test; after that every remaining group still fits in training, so validation is never reached. The split came out 0.50/0.00/0.50 instead of 0.60/0.20/0.20, which silently disabled early stopping and left half the dataset in test.

Nothing raised an error. The benchmark ran, produced plausible RMSEs, and wrote them to disk. The bug surfaced only because an unrelated experiment asked the splitter for a calibration set and got an empty array. Oversized groups are now assigned largest-first, so the group too large for any held-out subset inflates training rather than test, and a rebalancing pass moves whole groups into an undersized validation set. FreeSolv now splits 0.56/0.20/0.24. Results computed before the fix are not comparable with results after it. The MoleculeNet benchmark in Section 3.2 is therefore being regenerated under the corrected splitter; the pre-fix outputs are kept in `results/molnet_presplitfix/` so the two can be compared rather than quietly replaced.

4.2 A generator that lost to its own control

The graph diffusion model scored 0.0% validity — and the untrained marginal baseline, which samples atoms and bonds from the dataset frequencies with no model at all, scored higher. A trained model losing to its own control

sampler	valid	largest frag.	degree
model + valence	29.2%	100.0%	1.86
model, unconstrained	0.0%	0.0%	4.87
marginal baseline	0.0%	45.8%	1.83
training set	—	—	1.99

Table 1: Generation. Two validity conventions are reported because the literature uses both: a single connected valence-correct molecule, and the largest connected fragment. Mean degree against the training mean is the diagnostic that exposed the collapse.

is not a weak result; it is a broken one, and it was traced rather than written up.

Two bugs, compounding. "No bond" is 83.7% of atom pairs in a molecular graph, so under an unweighted cross entropy, predicting "none" everywhere is close to optimal; the denoiser had learned exactly that, producing graphs of mean degree 0.26 against a training-set mean of 1.99. Separately, the valence-constrained sampler began reverse diffusion from the all-zero bond matrix rather than the marginal noise prior, and since the reverse posterior's likelihood term anchors x_{t-1} to x_t, starting empty dragged the entire trajectory back towards empty.

With a class-balanced bond loss (exponent 1.5, bond weight 3.0) and prior-initialised sampling, mean generated degree rises to 1.86 and strict validity to 29.2%, against 0.0% for the marginal baseline (Table 1). Sampling the same trained model *without* the valence constraint gives 0.0% at mean degree 4.87, so the constraint is doing real work rather than masking a weak denoiser.

What remains wrong is specific: 70.8% of samples are rejected for being in two or more pieces, against 0.0% for valence errors. The model has no connectivity prior and the training objective supplies none. That is the next thing to fix, and it is stated here rather than hidden behind the lenient fragment-validity convention. These hyperparameters were selected on the validity metric reported above, with no held-out set, so that figure should be read as an upper bound.

4.3 An improvement that did not improve

The standard construction for adaptive conformal intervals [14] fits a separate model to the residuals and uses its prediction as the difficulty estimate. Implemented here — an inner fold held out of training, a second Gaussian process regressed on $\log|r|$ — it is worse than the constant-width baseline on every dataset tried: conditional-coverage drift of 0.321 on ESOL, 0.294 on Lipo and 0.219 on FreeSolv, against 0.111, 0.176 and 0.020 for doing nothing at all, with wider intervals throughout. It is reported, and not shipped.



Figure 4: The panel against the live service. Descriptors, scaffold families and the map colouring are computed by the NumPy package described here.

5 Integration

The pipeline ships as a panel for Mantis, a map interface for exploring large collections. The panel reads the structures behind selected points, computes descriptors and scaffolds, colours the map by any of them, proposes the next batch to assay, and returns predictions with conformal intervals. No file in the Mantis repository is modified: three files are added and three short registrations are documented as exact diffs. Typechecked against the real Mantis tree with its own dependencies and path aliases, the panel contributes zero errors.

6 Limitations

- Every implementation here is slower than its established counterpart, often by a large factor. Nothing in this work is a performance claim.
- The aromaticity and valence model is a documented subset, not a complete perception engine. Molecules outside it are reported invalid rather than guessed at.
- The generator produces disconnected graphs in the large majority of samples. Fragment-validity hides this; strict validity does not, which is why both are reported.
- Binding affinity is evaluated on one kinase panel. Cold-drug generalisation is measured, but on a single assay family.
- Generation hyperparameters were selected on the reported metric with no held-out set.
- Conformal guarantees are marginal. Conditional coverage is measured here because the marginal guarantee alone can disguise systematic over-confidence on the hardest cases, but no method here guarantees it.

7 Conclusion

A dependency-free reimplementation is not a contribution in itself, and this paper does not argue otherwise. What the exercise produced was a set of checks tight enough to catch three failures that would otherwise have passed silently: a split that emptied a validation set, a generator beaten by its own control, and an interval that met its guarantee while carrying no information. Each was found by comparing a number against something external — a published table, an untrained baseline, an analytically known answer — rather than by reading code. That comparison is cheap, and it is the part of this work that generalises.

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